Type MS · Run-length encoding (RLE)

Answer 1 of 4 · Question 1 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

1(ai)	0100 1101 100 1101 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(ii)	1000 0010 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(iii)	Award **one** mark for each of: • C (1) • 6 (1)	Must be in the correct order	2

Type MS · Run-length encoding (RLE)

Answer 1 of 4 · Question 1 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	400 pixels x 200 pixels x 12 bits / 8 bits per byte Allow expression in word form $\frac{400 \times 200 \times 12}{8}$ 400 x 200 **OR** 80000(1) 12/8 **OR** 1.5 (1)	Accept any other use of 400, 200, 12, 8 that gives the correct answer Allow one mark for 120,000 with no expression	2
1(b)(ii)	Spaces are to help legibility. Award **three** marks for: Line 1 3W 1B 4W Line 2 1W 6B 1W Line 3 1B 2W 1B 1W 1B 2W Line 4 3W 2B 3W Award **two** marks for any three lines correctly encoded Award **one** mark for any two lines correctly encoded	Allow **one** error in the sequence Accept letter before number but must be consistently used	3
1(b)(iii)	Award **one** mark from: • None of the original (image) detail/quality is lost (when the image is stored) (1) • None of the data/information is (permanently) removed (1) • It can be decompressed without losing detail (1)		1

Type MS · Run-length encoding (RLE)

Answer 1 of 4 · Question 1 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award up to **three** marks for a linked description: • set the sample rate/parameters/bit-depth (1) • sample (the analogue sound) (1) • measure the sound amplitude/volume/frequency (1) • give a (binary) value/number for each measurement (1) • store data as sample rate and values / digital signals (1)		3
1(c)(ii)	Award **one** mark from: • sound may be missing some frequencies (1) • some audio information/data is lost/deleted during the (compression) process (1)		1
		Total	**16**

Type MS · Run-length encoding (RLE)

Answer 1 of 4 · Question 1 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Run-length encoding (RLE)

Answer 2 of 4 · Question 2 part f.iii · 1 mark

2021 JUN · Paper 01 · extract 1 of 5 · find part f.iii in this mark scheme extract

Number			
2(a)	Award **one** mark from: • share peripherals/devices (e.g. printers, scanners) (1) • communicate (e.g. email, instant messaging, play games) (1) • share data (e.g. files/music/videos/backups on servers) (1) • deploy/update applications/software(1) • administrative purposes (e.g. remote access/desktop)		1

Question Number	Answer	Additional Guidance	Mark
2(b)	Award **one** mark from: • to allow (connected) machines to communicate (1) • to provide the rules of communication between (networked) devices (1)	Accept any other response indicating protocols enable communication between devices	1

Type MS · Run-length encoding (RLE)

Answer 2 of 4 · Question 2 part f.iii · 1 mark

2021 JUN · Paper 01 · extract 2 of 5 · find part f.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	**The only correct answer is A** ***B*** *is not correct because FTP is an application protocol, which is added in the application layer* ***C*** *is not correct because FTP is an application protocol, which is added in the application layer* ***D*** *is not correct because FTP is an application protocol, which is added in the application layer*		1
2(c)(ii)	Award **one** mark from: • Adds the source/destination IP address (1) • Establishes sockets (an IP address followed by a port number) (1) • Moves packets onto the next network node (1) • Strips the source/destination IP addresses (when it arrives at its destination) (1) • Passes the packet to the transport/next layer (when it arrives at its destination) (1) • Adds/removes packet headers (1)		1

Question Number	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for each of: **Domain Name** / **IP Address** / **URL** (blank) / (blank) / ✓ (blank) / ✓ / (blank) ✓ / (blank) / (blank)		3

Type MS · Run-length encoding (RLE)

Answer 2 of 4 · Question 2 part f.iii · 1 mark

2021 JUN · Paper 01 · extract 3 of 5 · find part f.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)(i)	Bus		1
2(e)(ii)	Award **one** mark from: • Easy to connect/remove devices/computers (in a network) (1) • Cable cost less / fewer cables needed (compared to mesh/star topology) (1) • Easier to plan/setup/maintain (1)		1
2(e)(iii)	Award **one** mark from: • If the main cable/a terminator fails or gets damaged the whole network will fail (1) • The more workstations the slower it gets (1) • The more workstations the more collisions (1) • Every device 'sees' all of the data on the network (1)		1

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	Award **two** marks from: • Reduces time taken to upload/download (1) • Reduces the bandwidth required/cost of data to upload/download (1) • (Usually) produces smaller file/reduces storage (on the web server/device) (1) • People have a limited hearing range / can't tell the difference (1) • People's brains will 'fill in the blanks' (1)		2
2(f)(ii)	**The only correct answer is C** ***A*** *is not correct because it is not compressed* ***B*** *is not correct because it uses lossless compression* ***D*** *is not correct because it uses lossless compression*		1
2(f)(iii)	Award **one** mark from: • 3c3w2c6w3c (1) • c3w3c2w6c3 (1)		1

Type MS · Run-length encoding (RLE)

Answer 2 of 4 · Question 2 part f.iii · 1 mark

2021 JUN · Paper 01 · extract 4 of 5 · find part f.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(g)	Award **four** marks for a correct expression or correct answer **Examples of correct expressions:** 6 x 1000 x 1000 x 1000 x 8 (4) 54 x 1000 x 1000 1000 x 8 (4) 9 888.9 (4) For an incorrect or partial expression award **one** mark for each of: • **Bits to transfer** ○ 6×1000^3 (1) ○ x 8 (or equivalent) (1) • **Speed in bits per second** ○ 54×1000^2 (1) • Their bits divided by their speed (1)	• Units are not required • Equivalent expressions are accepted • Allow follow through for incorrect or partial expressions	4

Type MS · Run-length encoding (RLE)

Answer 2 of 4 · Question 2 part f.iii · 1 mark

2021 JUN · Paper 01 · extract 5 of 5 · find part f.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Run-length encoding (RLE)

Answer 3 of 4 · Question 3 part e · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part e in this mark scheme extract

3(a)	Award up to **two** marks for a linked explanation, such as: • If the data is incomplete/inaccurate (1) the answers from the model might not be right (1) • If the assumptions the model is based on are inaccurate (1) the answers from the model might be incomplete/inaccurate (1)	Award one mark for identifying a problem and one mark for a reason There must be a clear cause and consequence/effect	2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for any of: • Pixels per inch/ppi (1) • Pixels per centimetre/ppc (1) • Dots per inch/dpi (1)	Ignore units cm^2 $inches^2$	1

Type MS · Run-length encoding (RLE)

Answer 3 of 4 · Question 3 part e · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part e in this mark scheme extract

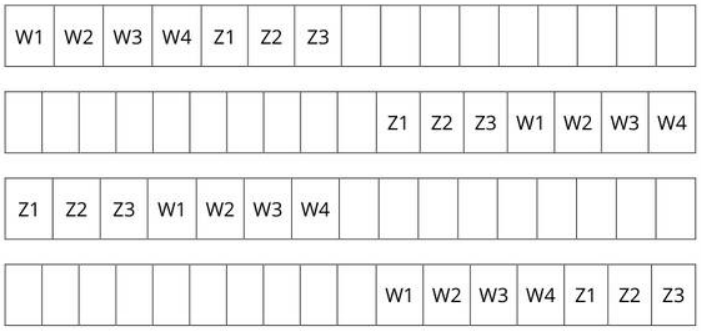
Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	Award **two** marks for a linked description such as: • Each pixel/colour in the image (1) will require more bits to store (1)	Giving for formula file_size = width*height*colour_depth on its own is not enough	2

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked description such as: • System software is used to maintain/run the operation of the system/hardware (1) whereas application software allows the user to carry out a task (1)	For both marks both categories must be addressed Allow clear examples	2

Type MS · Run-length encoding (RLE)

Answer 3 of 4 · Question 3 part e · 2 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part e in this mark scheme extract

Question number	Answer	Additional guidance	Mark
3(d)	Award **one** mark for each of: • All file blocks of each file are in numeric sequence/one after another in numeric order (1) • All blank space is together at the front/at the end (1) Examples: 	Note: Order of letter and number does not matter. W1 or 1W are equally acceptable	2

Type MS · Run-length encoding (RLE)

Answer 3 of 4 · Question 3 part e · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)	Award up to **two** marks for a linked explanation such as: • The file size will not be decreased/will increase (1) because there are very few/no repeating patterns / because a run length of one would be added to each digit (1)		2

Question Number	Answer	Additional Guidance	Mark
3(f)	Award **one** mark for each correct cell:		4

Task	Translator
A guessing game that can be used on different computing platforms	Interpreter (1)
A screen driver for a new smartphone	Assembler (1)
A new version of a spreadsheet program for sale next year	Compiler (1)
Control software for an embedded system inside a new washing machine	Assembler (1)

Type MS · Run-length encoding (RLE)

Answer 3 of 4 · Question 3 part e · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Run-length encoding (RLE)

Answer 4 of 4 · Question 3 part b.iii · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part b.iii in this mark scheme extract

3(a)(i)	Award **one** mark for each correct hexadecimal character in the correct order: • B9		**2**
Question	Answer	Additional Guidance	Mark
3(a)(ii)	**The only correct answer is B** *A is not correct because both are processed at the same speed by a computer* *C is not correct because both use the same amount of storage* *D is not correct because hexadecimal uses number base 16*		**1**
Question	Answer	Additional Guidance	Mark
3(a)(iii)	Award **two** marks for a linked description such as: • (logical/arithmetic) right shift (1) by 2 places (1)		**2**
Question	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for: Pixel/picture element (1)	Accept additional words. Accept incorrect spelling if word is recognisable as pixel	**1**

Type MS · Run-length encoding (RLE)

Answer 4 of 4 · Question 3 part b.iii · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part b.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
3(b)(ii)	Award **one** mark for each of: • 16x16 pixels (1) • Multiplying (16x16) by 2 bits per pixel in the numerator for colour depth (1) • Dividing by 8 in the denominator to convert to bytes (1) • Dividing 1000 by the number of bytes used for one image (1) Example: $1000 / \frac{16 \times 16 \times 2}{8}$ bytes Allow follow-through	4 marks for any of: 1000 X 8 / 16 x 16 x 2 8000 / 256 x 2 1000 / 16 x 2 x 2 1000 / 16 x 4 8000 / 512	4
Question	Answer	Additional Guidance	Mark
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • high compression ratio (1) repeated/continuous runs (of the same pixel colour) (1) • requires less storage space (1) encoding 16W for line 1 (1)	Allow valid RLE examples from the image. Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
3(c)	Award **one** mark for each correct cell: Encrypted text / Plain text RED COT	Ignore capitalisation Must be correct spelling	2
		Total for question 3	14

Type MS · Run-length encoding (RLE)

Answer 4 of 4 · Question 3 part b.iii · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part b.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark